



CSE-476: Introduction to Natural Language Processing
Quiz 1

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

- Q1. In the BIO tagging scheme for NER, what does the “I-” prefix indicate?
- A. Beginning of an entity
 - B. Inside an entity
 - C. Outside any entity
 - D. Between two tokens
- Q2. Which model considers only the observed words and ignores hidden labels?
- A. It uses only words (observations) and ignores labels
 - B. It is identical to an HMM
 - C. It models transitions between labels but does not consider the words
 - D. It models emissions $P(\text{word} \mid \text{label})$
- Q3. In a Hidden Markov Model (HMM) for NER, a transition probability is:
- A. $P(\text{label} \mid \text{previous label})$
 - B. $P(\text{label} \mid \text{word})$
 - C. $P(\text{word} \mid \text{previous word})$
 - D. $P(\text{word} \mid \text{label})$
- Q4. Which of the following best describes syntactic function labeling like subject and object?:
- A. A task that identifies predicates in a sentence and labels roles of elements such as agent, theme, or location
 - B. A process that assigns grammatical roles like subject and object based on syntactic structure
 - C. A method that classifies named entities and links them to a knowledge base

- D.** A technique that labels words with their parts of speech and morphological features
- Q5. Which of the following best describes run-length encoding (RLE)?
- A.** A technique that compresses text by replacing consecutive repeated characters with a count and a symbol
 - B.** A tokenization method that iteratively merges the most frequent pair of symbols in a corpus
 - C.** A tokenization method that splits text based on whitespace and punctuation
 - D.** A character-level encoding scheme that assigns fixed-length binary codes to characters
- Q6. Which of the following best describes the main goal of part-of-speech tagging in NLP?
- A.** To label each token with a grammatical category
 - B.** To convert text into a sequence of symbols suitable for modeling
 - C.** To reduce vocabulary size to zero
 - D.** To assign probabilities to sentences
- Q7. Why can splitting text by spaces lead to too many short tokens in some preprocessing setups?
- A.** It produces too many tokens
 - B.** It does not work well for languages without explicit word boundaries
 - C.** It cannot tokenize English at all
 - D.** It ignores punctuation entirely
- Q8. Which of the following is listed as a common sequence tagging task involving POS tags?
- A.** Part-of-speech tagging
 - B.** Named Entity Recognition
 - C.** Dependency parsing
 - D.** Sentiment analysis
- Q9. What is a common limitation of using too many distinct labels for NER?
- A.** It cannot represent overlapping or nested entities
 - B.** It requires too many labels
 - C.** It does not support multiple entity types
 - D.** It ignores entity boundaries
- Q10. When choosing between a fast but coarse model and a slow but precise model, which trade-off is most relevant?
- A.** Speed vs accuracy
 - B.** Tokenized sequence length vs vocabulary size

- C. Precision vs recall
- D. Memory vs computation